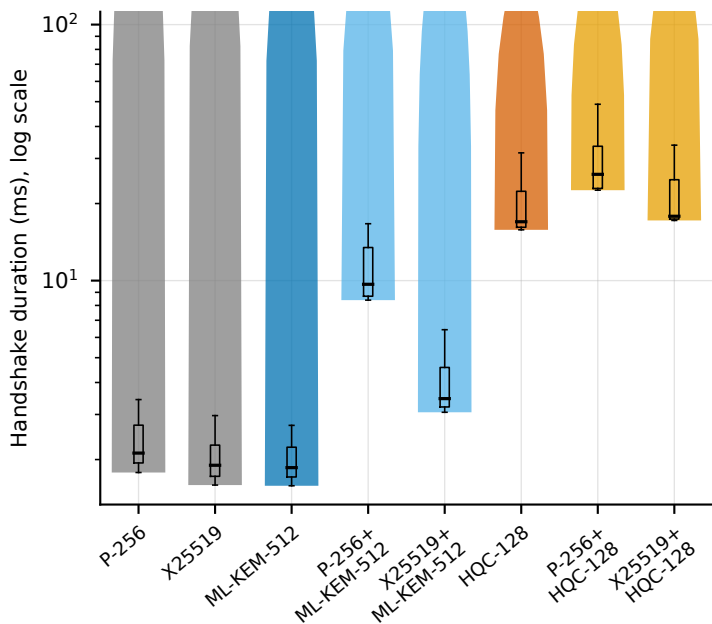


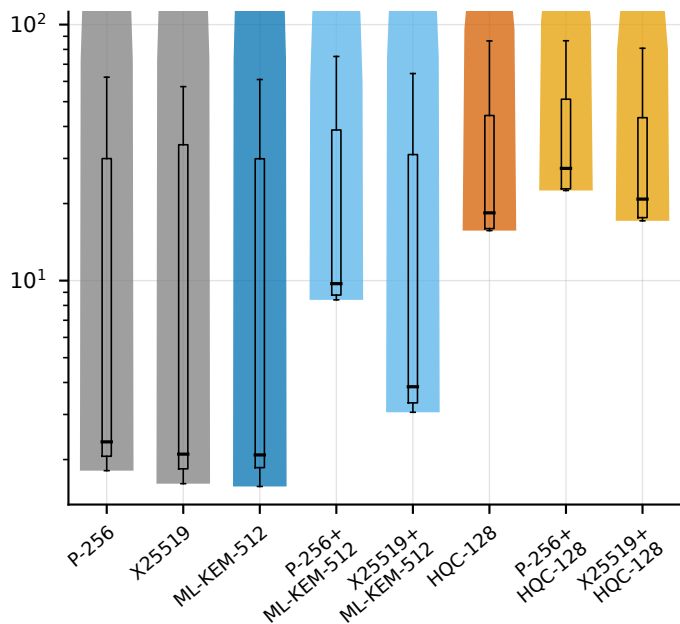
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	354.5	264.7	7.00k	1/500	342.3	271.3	7.02k	0
X25519	265.2	314.8	8.00k	6/500	402.7	241.3	7.10k	1/500
ML-KEM-512	253.2	275.1	7.00k	1/500	411.2	256.6	7.11k	0
P-256+ML-KEM-512	241.5	254.7	7.01k	0	283.1	246.7	5.04k	0
X25519+ML-KEM-512	256.1	219.2	3.01k	0	300.2	278.5	7.03k	0
HQC-128	83.3	390.4	3.04k	0	182.2	326.6	8.01k	0
P-256+HQC-128	112.7	331.5	3.04k	0	184.5	315.5	8.06k	0
X25519+HQC-128	140.0	346.5	7.01k	0	150.3	332.6	6.21k	0